

Question 1: Identify the content of R1 after the following instructions sequence is executed

MOV R1,#8

MOV R2,#3

ADD R3,R1,R2

SUB R2,R3,R1

MULT R4,R2,R2

ADD R2,R4,R1

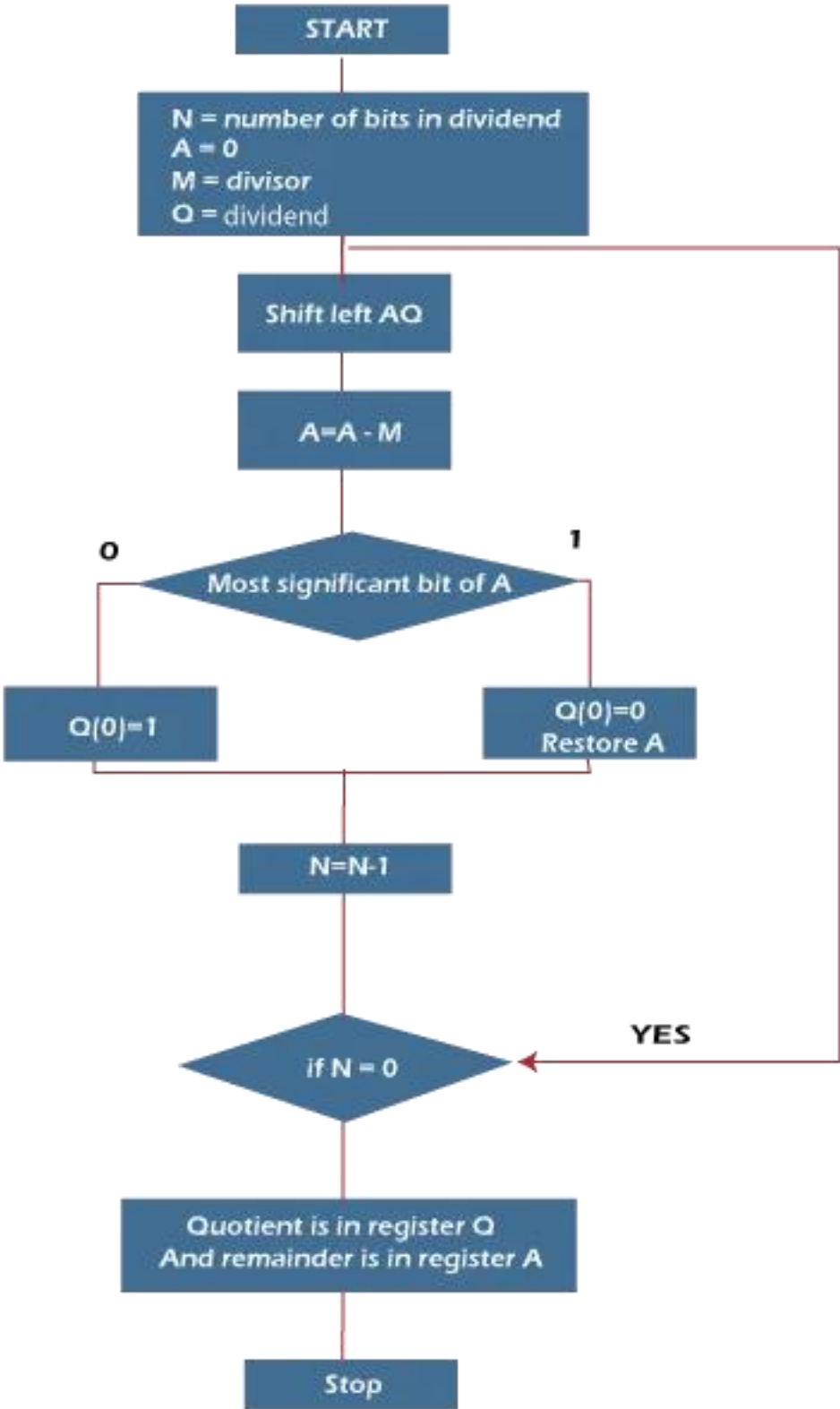
Instruction	Effect / Register values
MOV R1,#8	R1 = 8
MOV R2,#3	R2 = 3
ADD R3,R1,R2	$R3 = R1 + R2 = 8 + 3 = 11$
SUB R2,R3,R1	$R2 = R3 - R1 = 11 - 8 = 3$
MULT R4,R2,R2	$R4 = R2 \times R2 = 3 \times 3 = 9$
ADD R2,R4,R1	$R2 = R4 + R1 = 9 + 8 = 17$

R1 is never re-assigned after the first instruction, so it keeps its initial value.

Answer: R1 = 8 (binary 1000)

Question 2: Divide 1010 (10) by 0010 (2) – Restoring and Non-Restoring Division

(a) Restoring Division

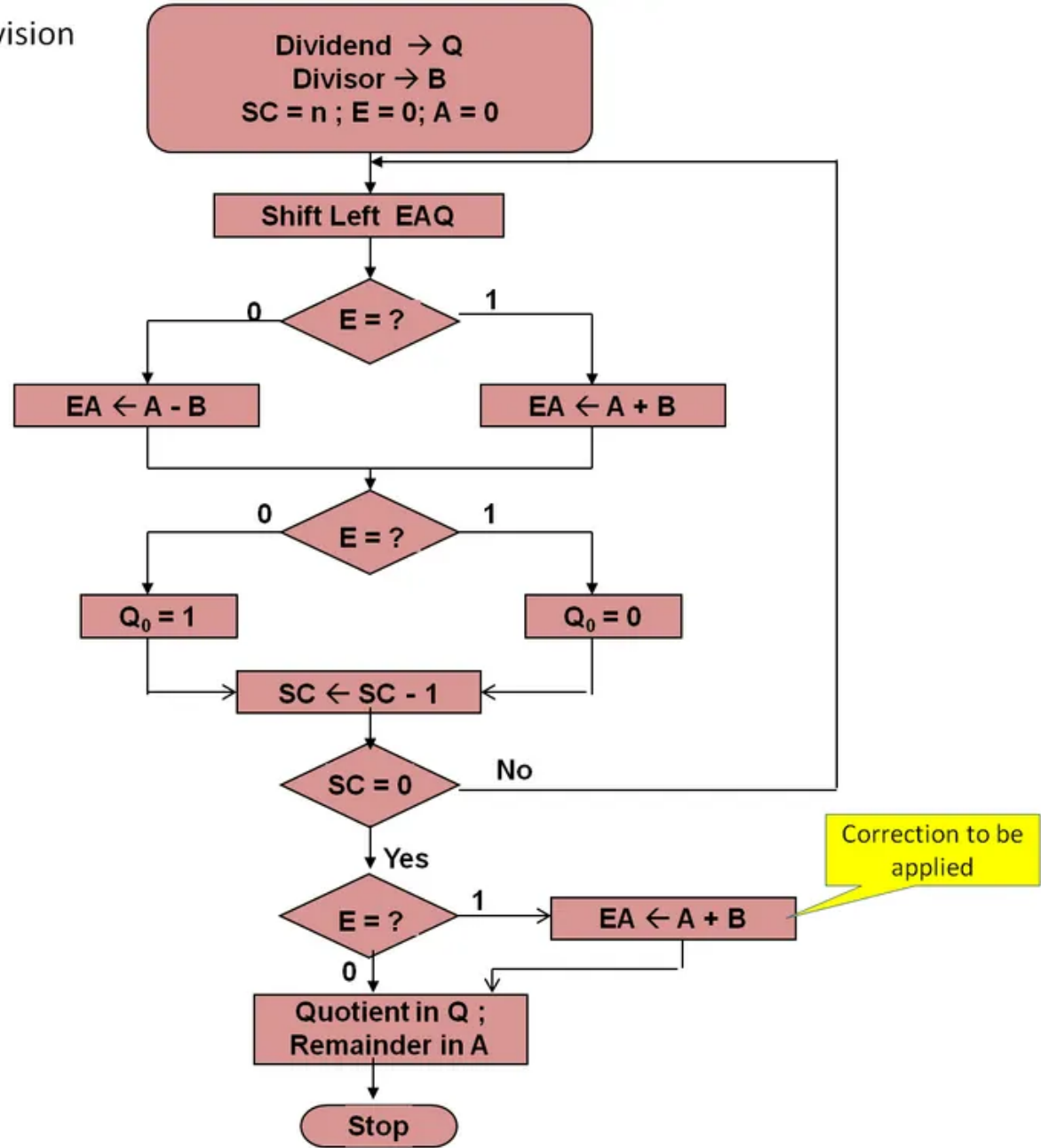


Step	A	Q	Operation
0	00000	1010	Initial: A = 0, Q = 1010, M = 0010
1	00001	0100	Shift AQ left → A=00001; A-M = 11111 (neg) → restore A=00001, Q0=0
2	00000	1001	Shift AQ left → A=00010; A-M = 00000 (≥0) → Q0=1
3	00001	0010	Shift AQ left → A=00001; A-M = 11111 (neg) → restore A=00001, Q0=0
4	00000	0101	Shift AQ left → A=00010; A-M = 00000 (≥0) → Q0=1

Result: Quotient Q = 0101 (5), Remainder A = 0000 (0)

(b) Non-Restoring Division

Non restoring division



Step	A	Q	Operation
0	00000	1010	Initial: A = 0, Q = 1010, M = 0010
1	11111	0100	A ≥ 0 → shift, A = A - M = 11111 (neg) → Q ₀ = 0
2	00000	1001	A < 0 → shift, A = A + M = 00000 (≥ 0) → Q ₀ = 1
3	11111	0010	A ≥ 0 → shift, A = A - M = 11111 (neg) → Q ₀ = 0
4	00000	0101	A < 0 → shift, A = A + M = 00000 (≥ 0) → Q ₀ = 1; A ≥ 0 → no final correction

Result: Quotient Q = 0101 (5), Remainder A = 0000 (0)

Advantages

- **Restoring division:** Simple to understand and implement; the accumulator always holds a valid non-negative partial remainder after every step, which makes the control logic straightforward.
- **Non-restoring division:** Faster and more hardware-efficient, since it performs only one add/subtract per cycle and avoids the extra restoring addition step, reducing the total number of operations.